

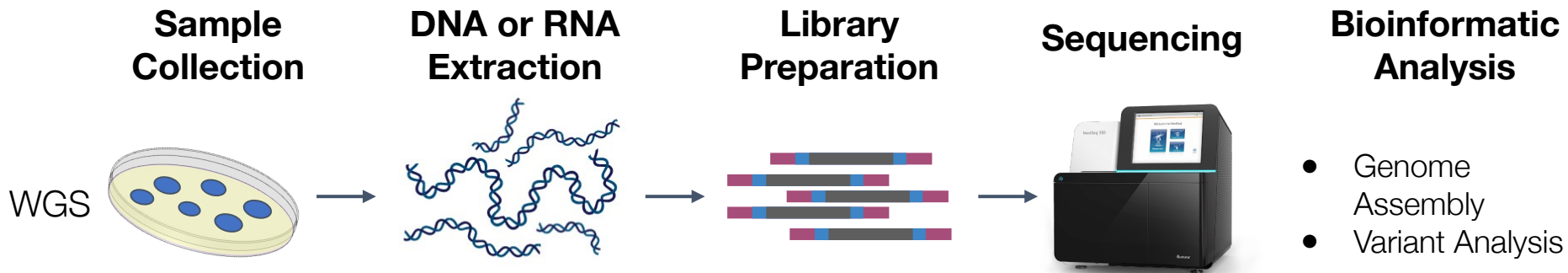
Bioinformatics Basics



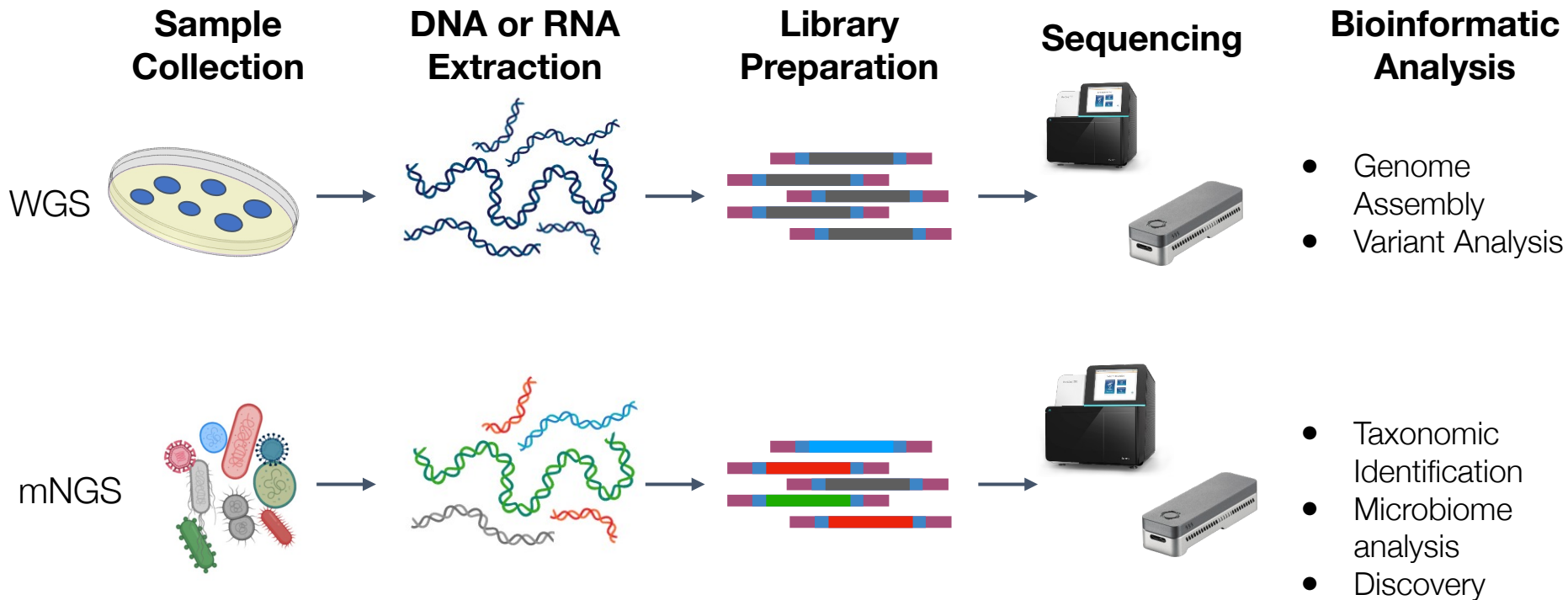
Outline

- Review NGS workflow
 - Whole genome sequencing (WGS) vs metagenomic NGS (mNGS)
- Sequence File Formats
 - FASTA vs FASTQ
- Pipeline basics
 - WGS workflow overview
 - mNGS common workflow steps
- BLAST

WGS vs mNGS Workflow



WGS vs mNGS Workflow Overview



Sequence File Format: FASTA vs FASTQ

- FASTA
 - Includes sequence name and sequence

Sequence Identifier → >Nucleotide Sequence Name

Sequence {
ATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGG
CTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATT
GGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCT

Sequence File Format: FASTA vs FASTQ

- FASTA
 - Includes sequence name and sequence

Sequence Identifier → >Nucleotide Sequence Name
Sequence { ATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGG
CTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATT
GGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCT

>Protein Sequence Name
MDITIHNPLIRRPLFSWLAPSRIFDQIFGEHLQESELLPASPSLSPFLM
RSPIFRMPSWLETGLSEMRLEKDKFSVNLDVKHFSPEELKVKVLGD
MVEIHGKHEERQDEHGFIAREFNRKYRIPADVVDPLTITSSLSLDGVL

Sequence File Format: FASTA vs FASTQ

- FASTA
 - Includes sequence name and sequence

>Nucleotide Sequence Name

```
ATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGG  
CTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATT  
GGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCT
```

- FASTQ
 - Includes sequence identifier, sequence, and **quality scores** for base calls

```
Sequence Identifier → @NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA  
Sequence → CATTCGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCCA  
+  
AAAAAEEEEEEEEEEEE/EEEEEEEEEEEEEEEEEEEEEEEE<EEEEEEEEEE//EAEEEEEEEAA/EEE/EEAEEEE<6/
```

Raw Sequence Reads: FASTQ format

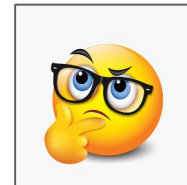
Example Read

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTCGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCCA
+
AAAAAEEEEEEEEEEEE/EEEEEEEEEEEEEEEEEEEE<EEEEEEEEEE//EAEEEEEEAA/EEE/EEAAEEEE<6/
```

1. A sequence identifier with information about the run.
2. The base calls: A, C, T, G and N.
3. A separator, which is simply a plus (+) sign.
4. The base call quality scores or Phred scores (coded in [ASCII format](#))

Quality Score	Probability of Incorrect Base Call	Inferred Base Call Accuracy
10 (Q10)	1 in 10 = 10%	90%
20 (Q20)	1 in 100 = 1%	99%
30 (Q30)	1 in 1000 = 0.1%	99.9%

$$Q = -10\log_{10}(e)$$



How many reads are listed here?

File: Sample_R1.fastq

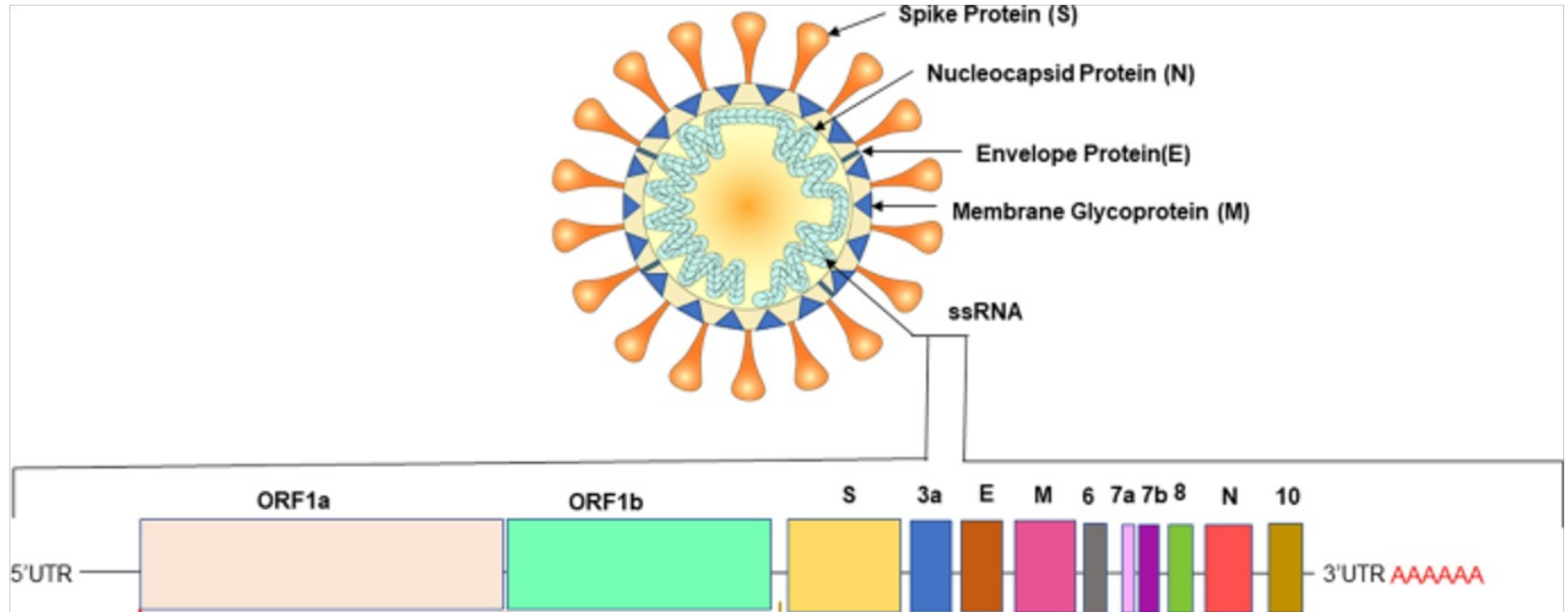
```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCATCTCGGCG
ATCAGTTTGATGTCTTCTTTGTAGCGACCGTAGAAGTCGACGGCTTCGTGGTTGCGGTTAGTATTTN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE<EEEEEEEE//EAEAAAAA/EEE/EEAAAAEE<6/<AEE<E<A//EEEEAAEE<EA/
<AA//EAAEAAE<E/EAEE///E<A/<AA//A<A<<A6<A//AA/E///6A#
@NB501961:14:HM7TLBGX2:1:11103:13923:13163 1:N:0:GATCACCA+GATCACCA
GAACTGGAGCTTTAGCCCGGTGATCCTGTGCCTCATTTGTATGGCTACCGCCATGGTGATGTACCTGCCATTTCTCAAAGCTTACG
AAAAACAGCTGCTGGCCAGGAGCGTGAAAACGCCGTCGGCCAGGCTGACAACGCGGCGCAGAN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE/
EEEEEEAAAEAAEAAEAAEAAAAAAAAAAAAAAAAAAAAEEEEEEEEEEEEEEEE<EEEEEEAE<AEAEAA<AAEAEAAE/6/A/EEEE/
EEE<AE/#
@NB501961:14:HM7TLBGX2:1:11104:15318:11334 1:N:0:GATCACCA+GATCACCA
GATAGTGCTGTTTCGCGAGCGATCGCCAGTACGCGCTCGCGCTCTCCTGCGCCACGCGCAGCTCTTACCACGACCGTTGAGTA
CCAGGCTGGCGGTGCTTTGGAGACGCCCGCCAGTTCGGCGATATCTTTGATAGTTACGCGTTTGG
+
AAAAEEEEEEEEAAEEAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEAAEEAEE<EAAE/EE/EEE<AAAAEE/
EEEEEEEEEEEE/EEE//AEE<E/EEEEAAE/A<<AEEEE/<66A///<</AA<//6AE<<6/<6
```

How do we get from here...

sample_R1.fastq

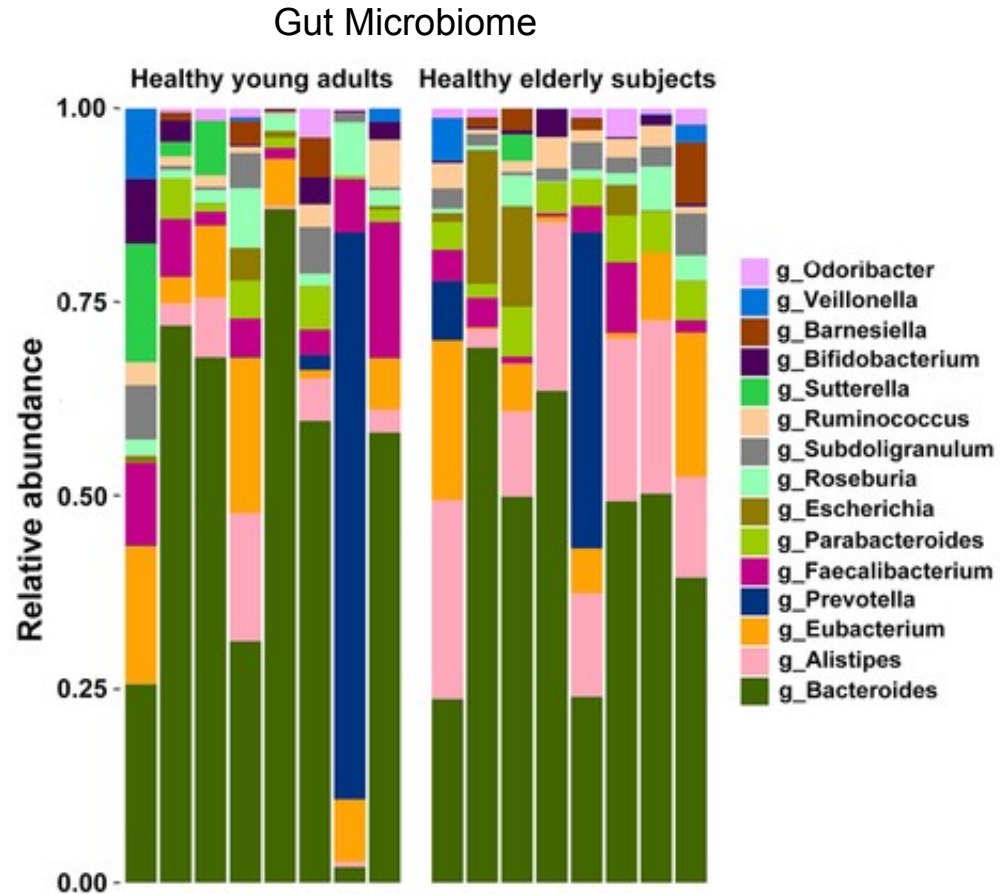
```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCATCTCGGCG
ATCAGTTTGATGTCTTCTTTGTAGCGACCGTAGAAGTCGACGGCTTCGTGTTTCGGGTAGTATTTN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE<EEEEEEEE//EAEAAAAA/EEE/EEAAAA<6/<AEE<E<A//EEEEAAEE<EA/
<AA//EAAEAAE<E/EAEE///E<A/<AA//A<A<<A6<A//AA/E///6A#
@NB501961:14:HM7TLBGX2:1:11103:13923:13163 1:N:0:GATCACCA+GATCACCA
GAACTGGAGCTTTAGCCCGGTGATCCTGTGCCTCATTTGTATGGCTACCGCCATGGTGATGTACCTGCCATTTCTCAAAGCTTACG
AAAAACAGCTGCTGGCCAGGAGCGTGAAAACGCCGTCGGCCAGGCTGACAACGCGGCGCAGAN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE/
EEEEEEAAAEAAEAAEAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAA<EEEEEEAE<AEAEAA<AAEAEAAE/6/A/EEEE/
EEE<AE/#
@NB501961:14:HM7TLBGX2:1:11104:15318:11334 1:N:0:GATCACCA+GATCACCA
GATAGTGCTGTTTCGCGAGCGATCGCCAGTACGCGCTCGCGCGTCTCCTGCGCCACGCGCAGCTCTTACCACGACCGTTGAGTA
CCAGGCTGGCGGTGCGCTTTGGAGACGCCCGCCAGTTCGGCGATATCTTTGATAGTTACGCGTTTGG
+
AAAAEEEEEEEEAAEEAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE
EEEEEEEEEEEE/EEE//AEE<E/EEEEAAE/A<<AEEEE/<66A///<</AA<//6AE<<6/<6
```

...to here?



[Rastogi et al., 2020](#)

...or here?

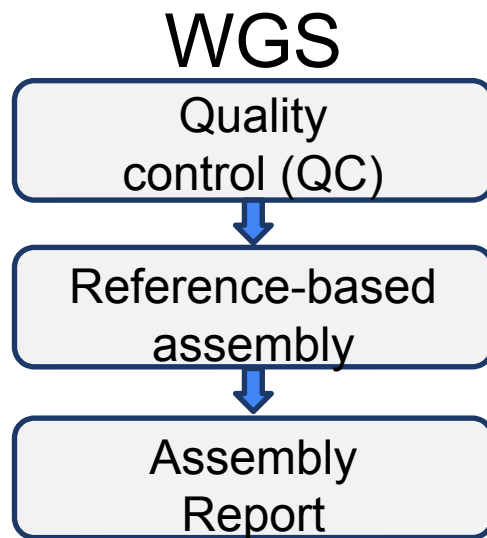


Bioinformatic Pipeline Basics

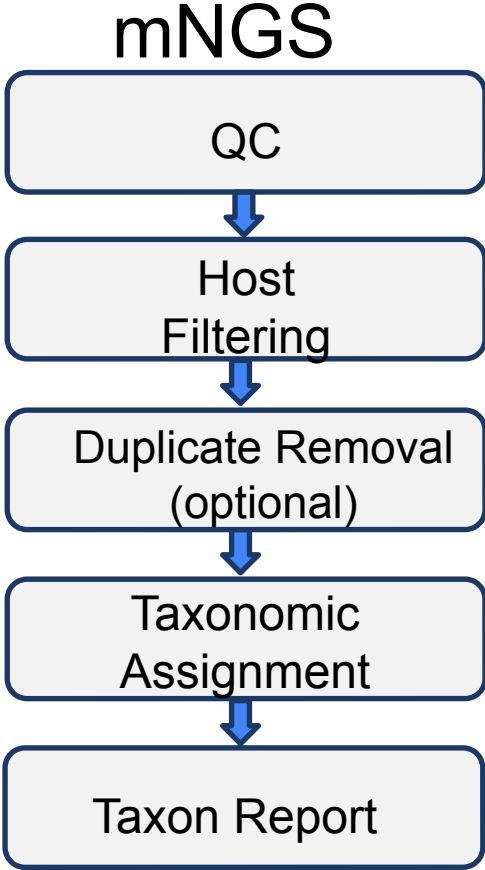
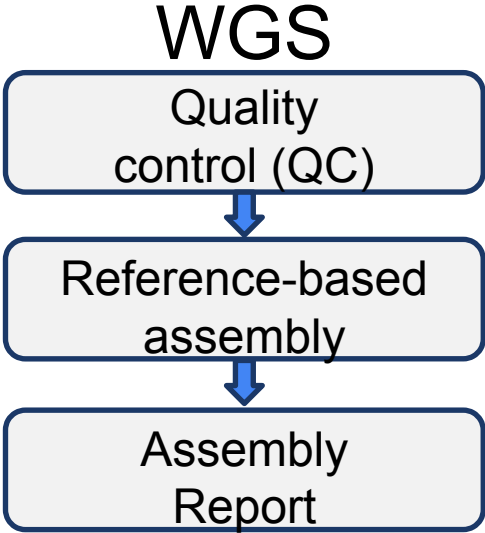
Pipeline Goals

WGS	mNGS
• Map reads against a reference genome sequence to obtain a consensus genome	• Assign sequencing reads to taxa by comparing sequences against reference databases • Assess relative abundance of taxa based on the number of sequences matching a given taxon

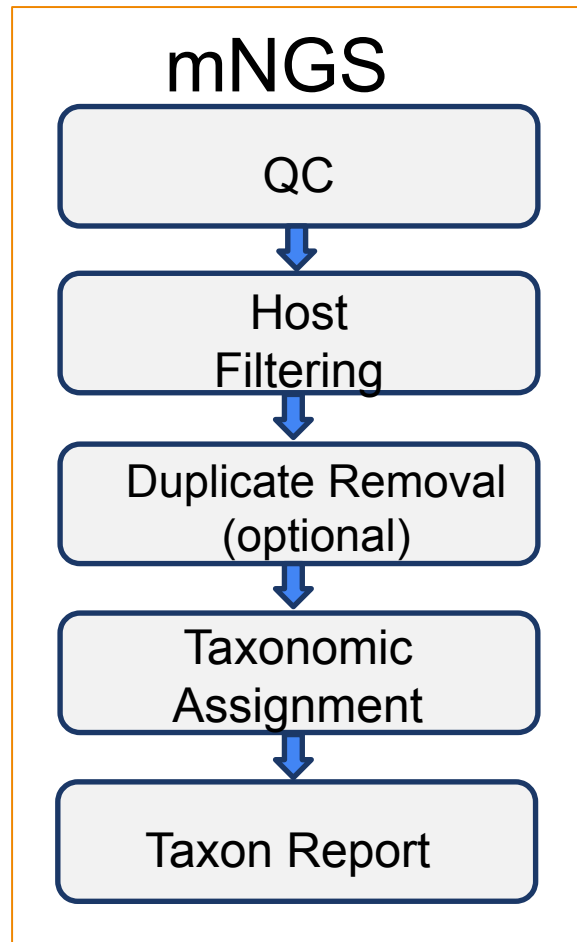
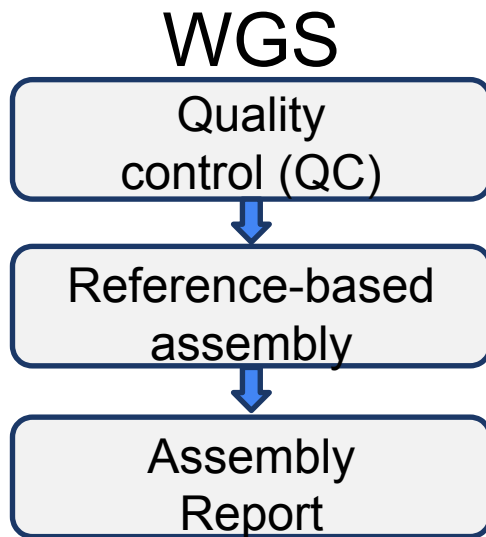
Common Pipeline Steps



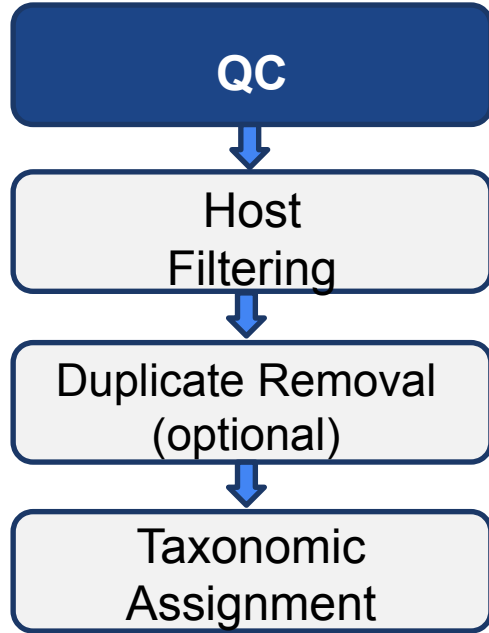
Common Pipeline Steps



Common Pipeline Steps



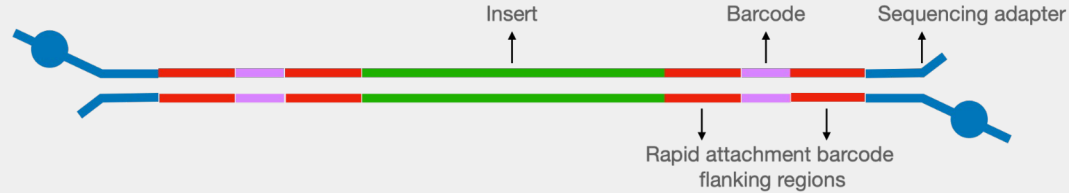
Common mNGS Pipeline Workflow



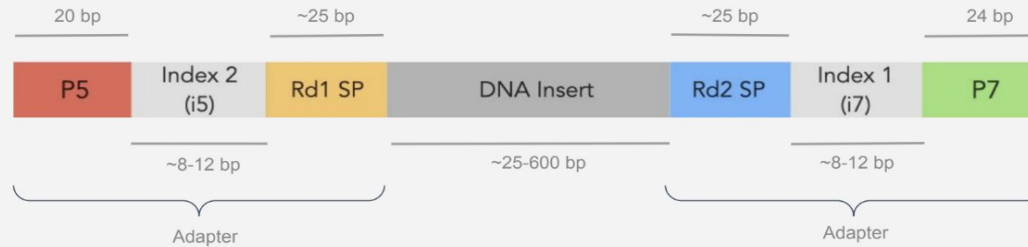
Goal: Remove bad quality and uninformative reads

Step 1: Quality Control (QC)

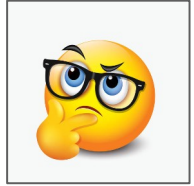
Anatomy of a standard ONT library



Anatomy of a standard Illumina library

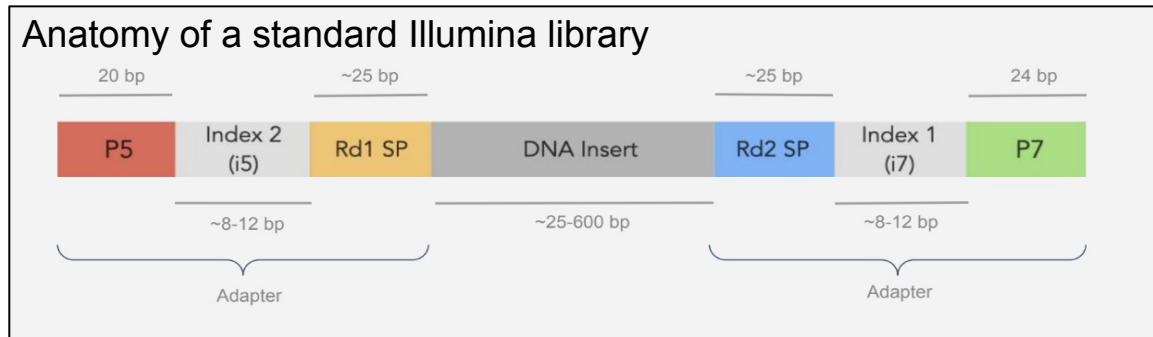
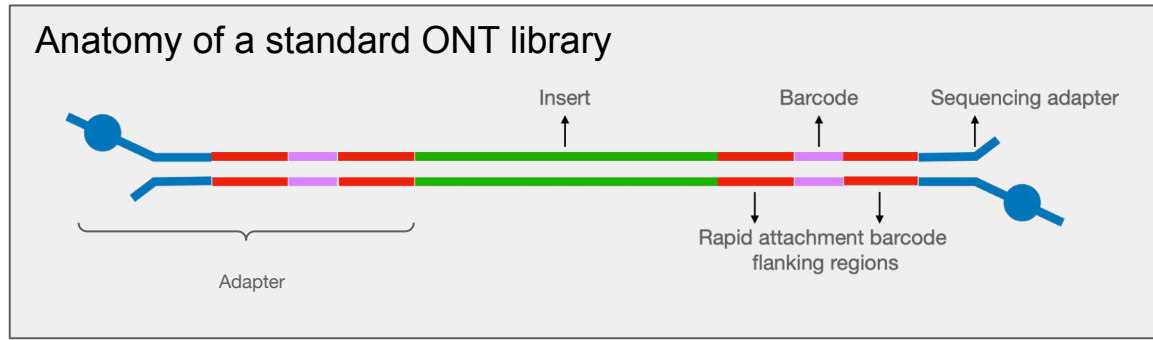


Step 1: Quality Control QC



Why?

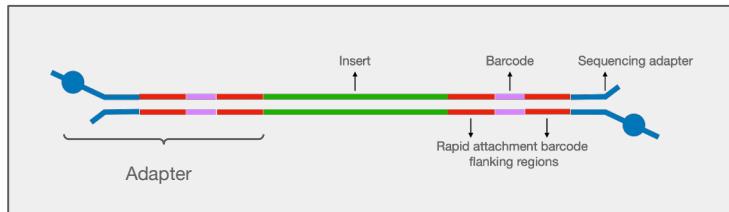
- Software to:
 - Trim adapters (synthetic, non-genomic)



Step 1: Quality Control QC

- Software to:
 - Trim adapters (synthetic, non-genomic)

Anatomy of a standard ONT library



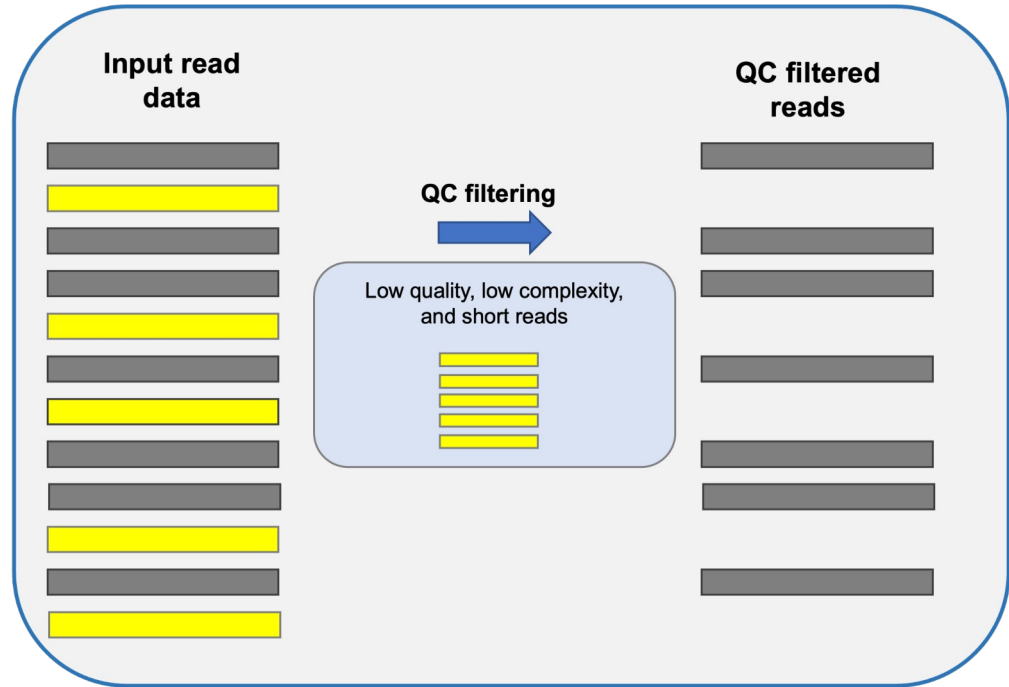
Anatomy of a standard Illumina library



- Filter reads based on quality, complexity, and length
 - Quality: Phred score threshold to ensure high quality data
 - Complexity: Remove low complexity, repetitive sequences (non-informative)
 - Length: Sequence length threshold to remove short sequences

Step 1: QC

Step: Trim adapters and filter reads based on quality (quality scores, complexity, and length)



Step 1: QC

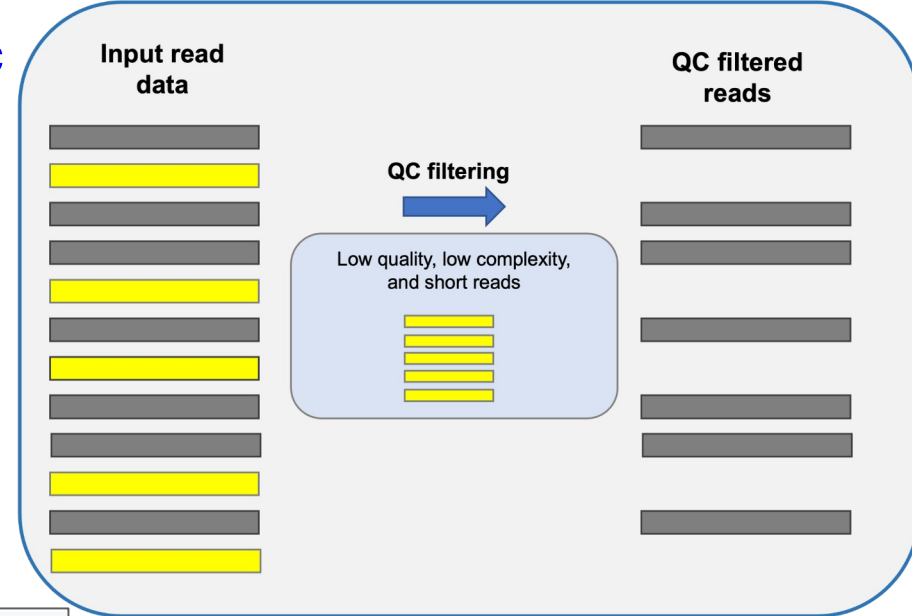
Example Software: [fastp](#), [Trimmomatic](#)

Example Parameters (fastp):

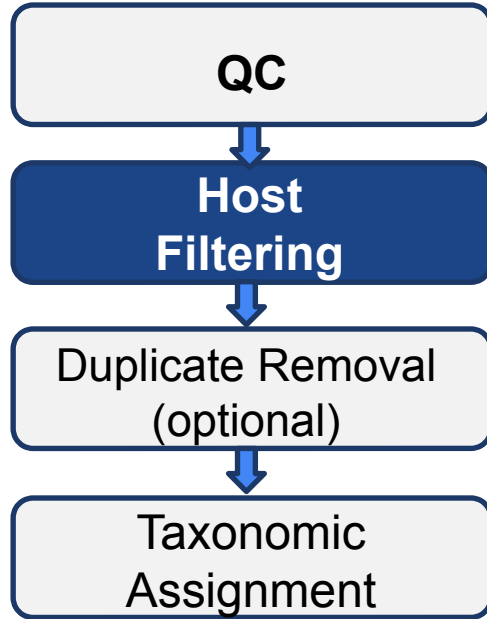
- **Qualified_quality_phred:** (i.e., bases with quality scores above a threshold*)
- **Unqualified_percent_limit:** % of the bases allowed to be unqualified
- **Min read length:** 35bp (Illumina) 100 (ONT)
- **Low_complexity_filter:** min % of bases followed by identical base

```
# a 51-bp sequence, with 3 bases that is different from its next base
seq = 'AAAAATTTTTTTTTTTTTTTTTTTTGGGGGGGGGGGGGGGGGGGGCCCC'
complexity = 3/(51-1) = 6%
```

- **N_base_limit:** 15 (i.e., discard reads containing > 15 Ns)

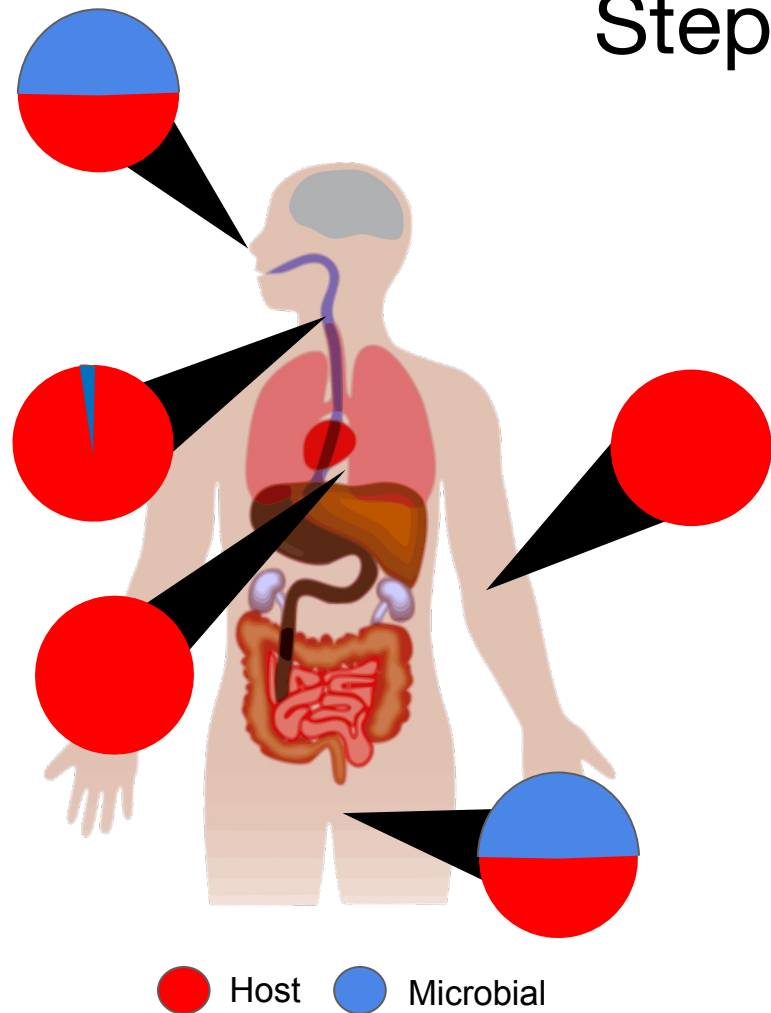


Common mNGS Pipeline Workflow



Goal: Filter out host reads to maximize the microbial signal, reduce false-positives, and protect private information

Step 2: Host Filtering

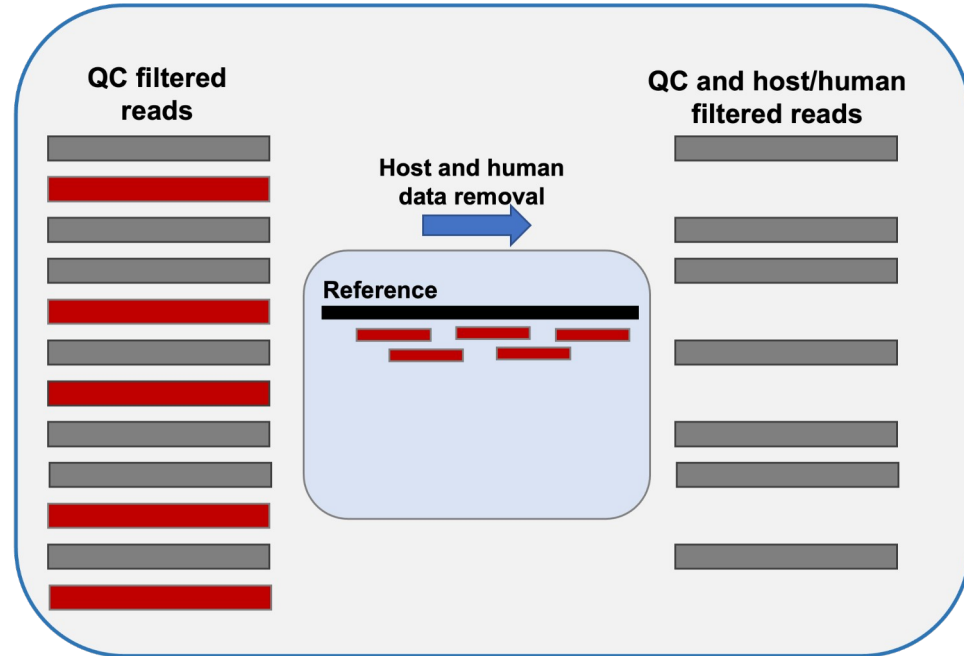


Data may be dominated by host sequences

Sample Type	Host Signal	Microbial Signal
CSF	> 99%	<< 5% with infection
Tracheal Aspirate	95%	5%
Nasal or oral	50%	50%
Whole blood	> 99%	<< 1% with infection
Rectal or Vaginal	50%	50%

Step 2: Host Filtering

Step: Host read removal through subtractive alignment against reference.

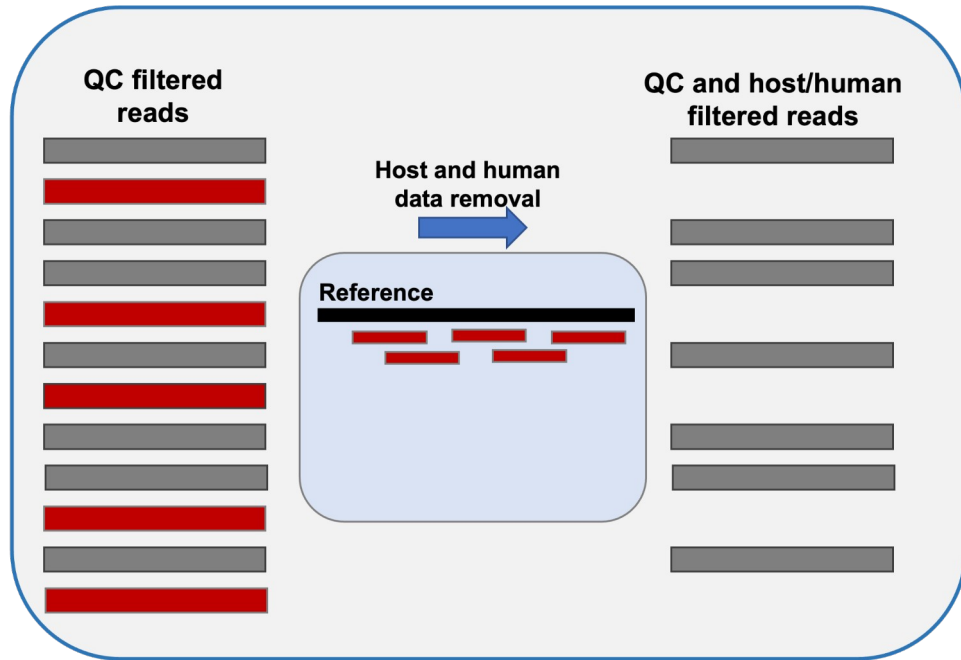


Step 2: Host Filtering

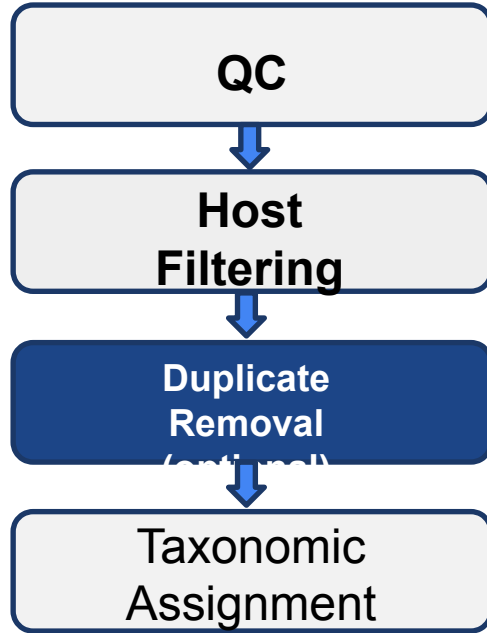
Example Software: [Bowtie2](#), [minimap2](#), [HISAT2](#)

Notes:

- **Human:** Reads mapped against [HG38](#) and/or [T2T-CH M13](#) reference genomes.
- **Non-human host:** Reads mapped against reference sequence.
- **Bowtie2:** Optimized for short reads (DNA)
- **HISAT2:** Optimized for short reads (DNA and RNA)
- **Minimap2:** Versatile (short- or long-reads, DNA or RNA)
- [Samtools](#) can be used to keep reads *NOT* mapping to the reference genome.



Common mNGS Pipeline Workflow



Goal: Cluster duplicate reads to lower computational burden (optional)

Step 3: Clustering duplicates

Software to:

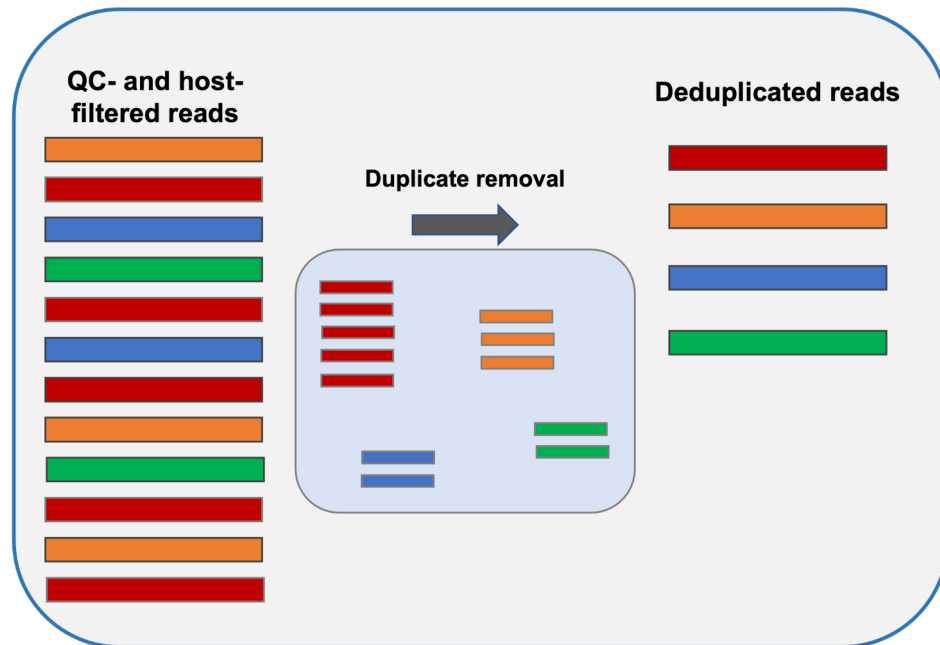
- Identify and cluster duplicate sequences (i.e., deduplication)

Clustering duplicates

Example Software: [CD-HIT-DUP](#)

Notes:

- Reads are clustered based on specified identity threshold and only representative reads from each cluster are used in downstream analysis.
- Cluster information can be retained to calculate non-duplicated values



Clustering duplicates

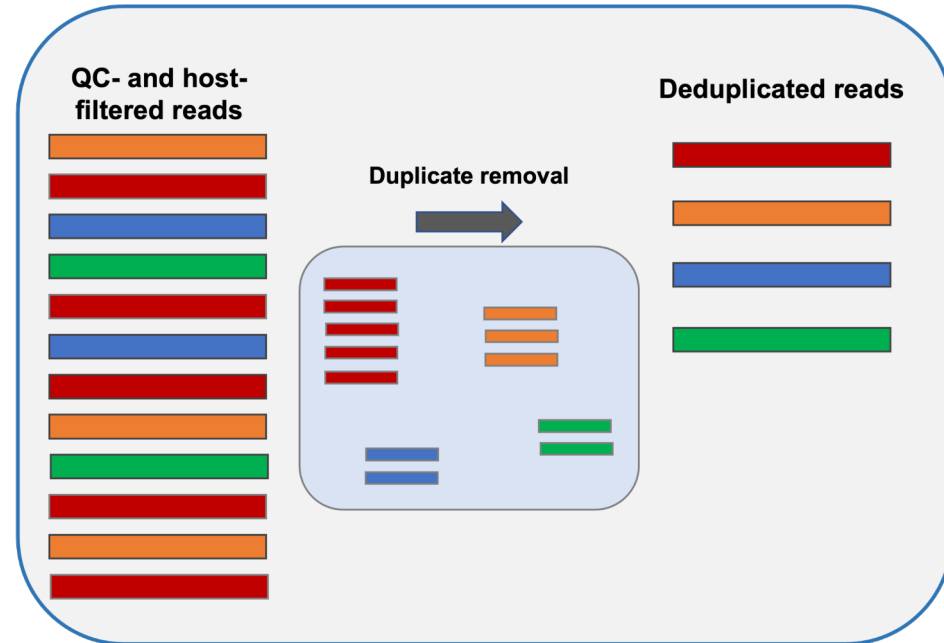


Why calculate non-deduplicated values?

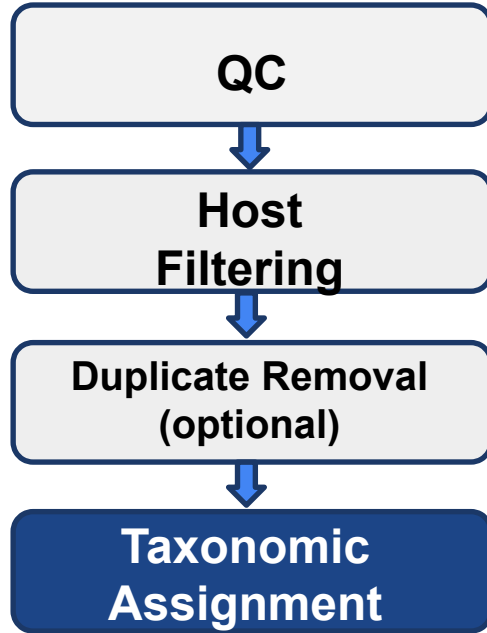
Example Software: [CD-HIT-DUP](#)

Notes:

- Reads are clustered based on specified identity threshold and only representative reads from each cluster are used in downstream analysis.
- Cluster information can be retained to calculate non-deduplicated values



Common mNGS Pipeline Workflow



Goal: Identify taxa and assess their abundance based on sequence comparisons against a reference database.

Taxonomic Assignment

Software to:

- Compare sequences against a reference database
 - k-mer based approaches
 - Alignment

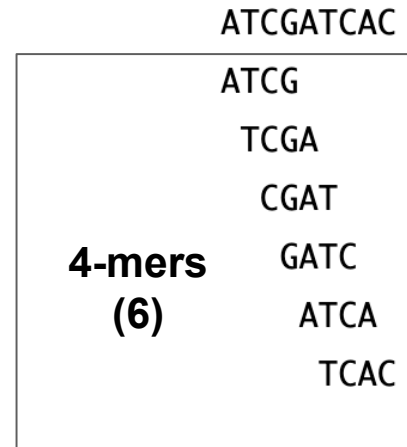
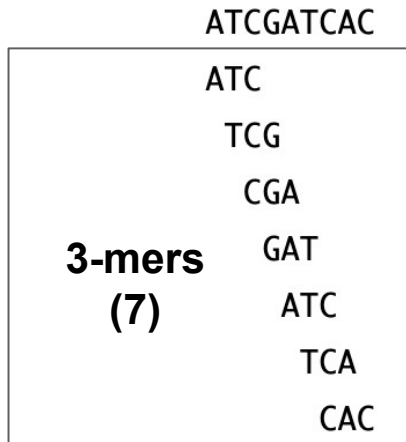
Taxonomic assignment: k-mer based approaches

Goal: Identify query sequences based on the presence of similar substrings (pieces of sequence) in species found in a reference database.

Taxonomic assignment: k-mer based approaches

Goal: Identify query sequences based on the presence of similar substrings (pieces of sequence) in species found in a reference database.

k-mer: Substring of length k in a given sequence. To get all k-mers from a sequence, get the first k characters and move just a single character to start the next k-mer and so on



Taxonomic assignment: k-mer based

Example: [Kraken2](#)

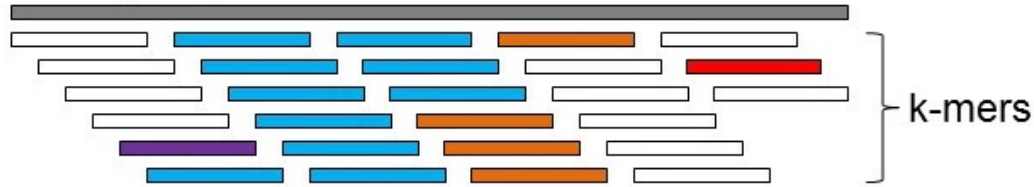
Kraken:

- Taxonomic classification system using **exact k-mer matches**.
- Matches each k-mer within a query sequence to the **lowest common ancestor (LCA)** of all genomes containing the given k-mer.

Taxonomic assignment: k-mer based

Example: [Kraken2](#)

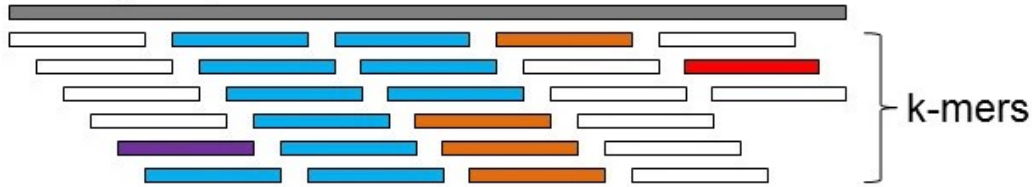
1. Extract k-mers from sequence read (31-mers)



Taxonomic assignment: k-mer based

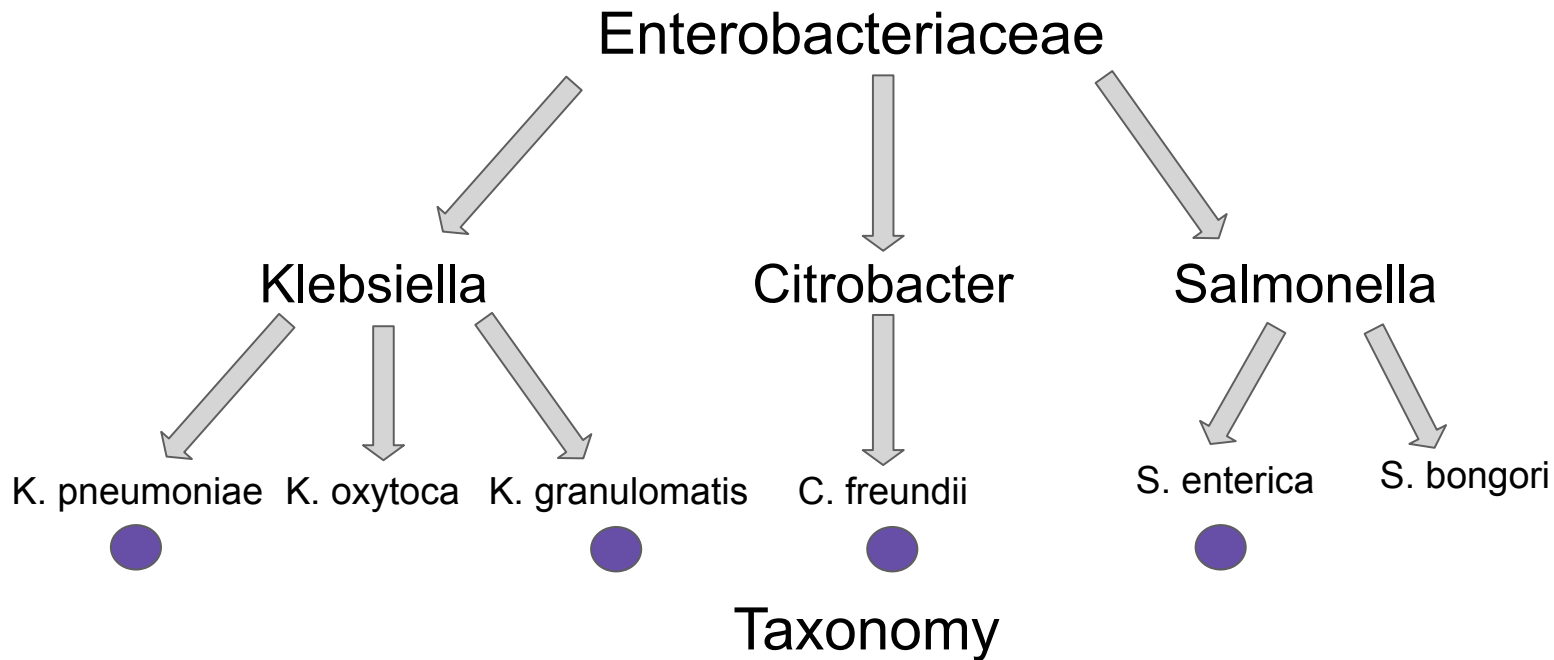
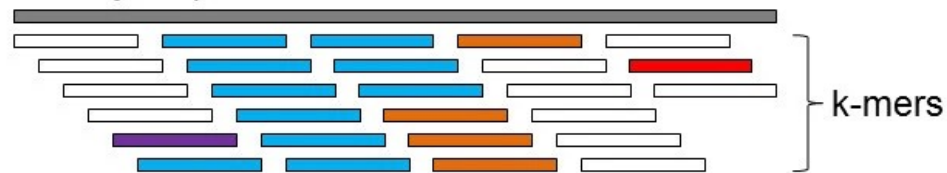
Example: [Kraken2](#)

1. Extract k-mers from sequence read (31-mers)

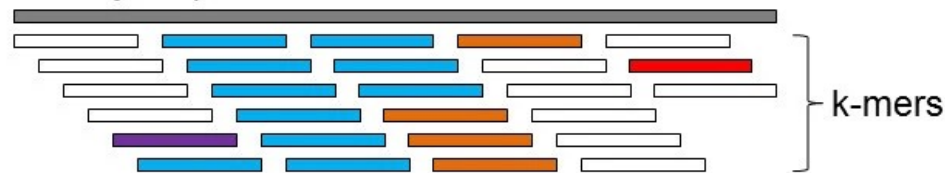


2. Compare to pre-computed database to determine taxonomic ID of reference sequences with specified k-mer (Lowest Common Ancestor, **LCA**, mapping)

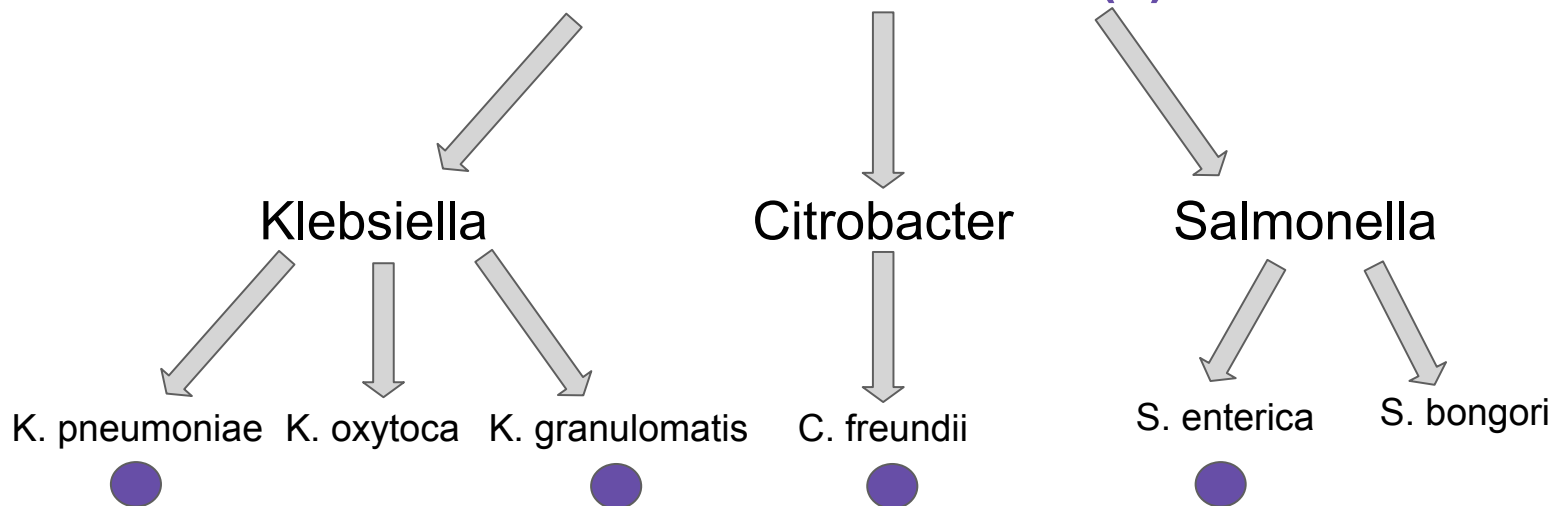
LCA Mapping



LCA Mapping

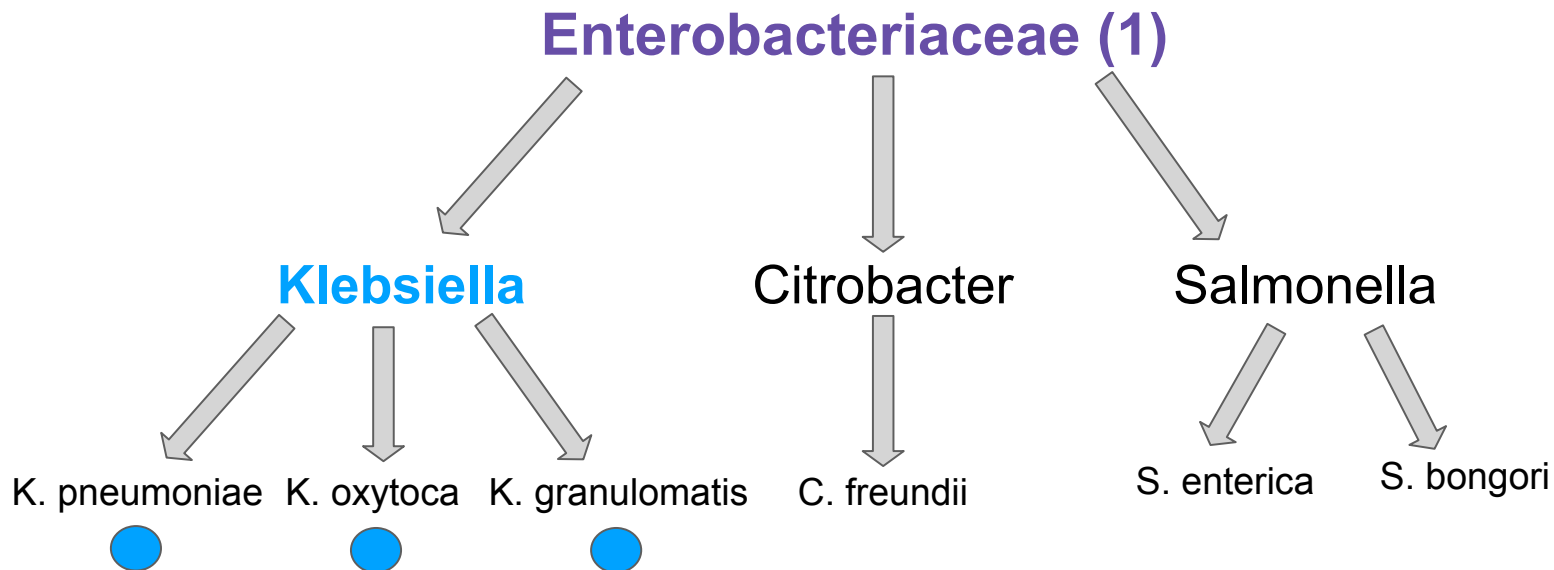
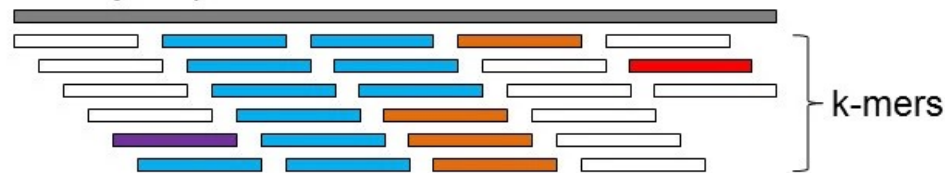


Enterobacteriaceae (1)



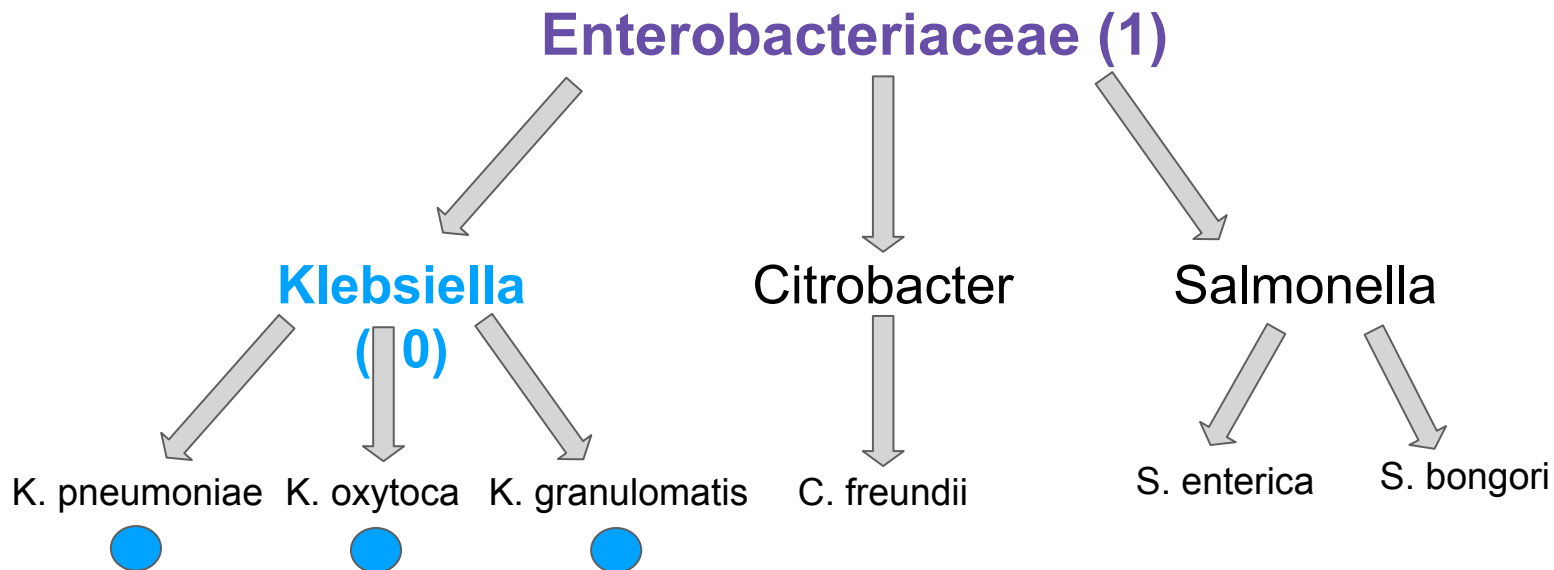
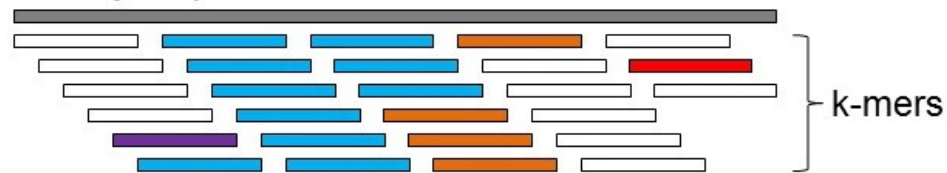
Taxonomy

LCA Mapping



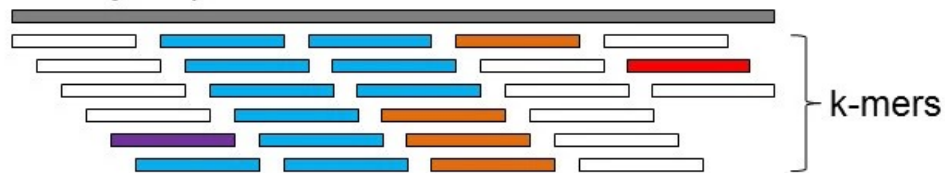
Taxonomy

LCA Mapping

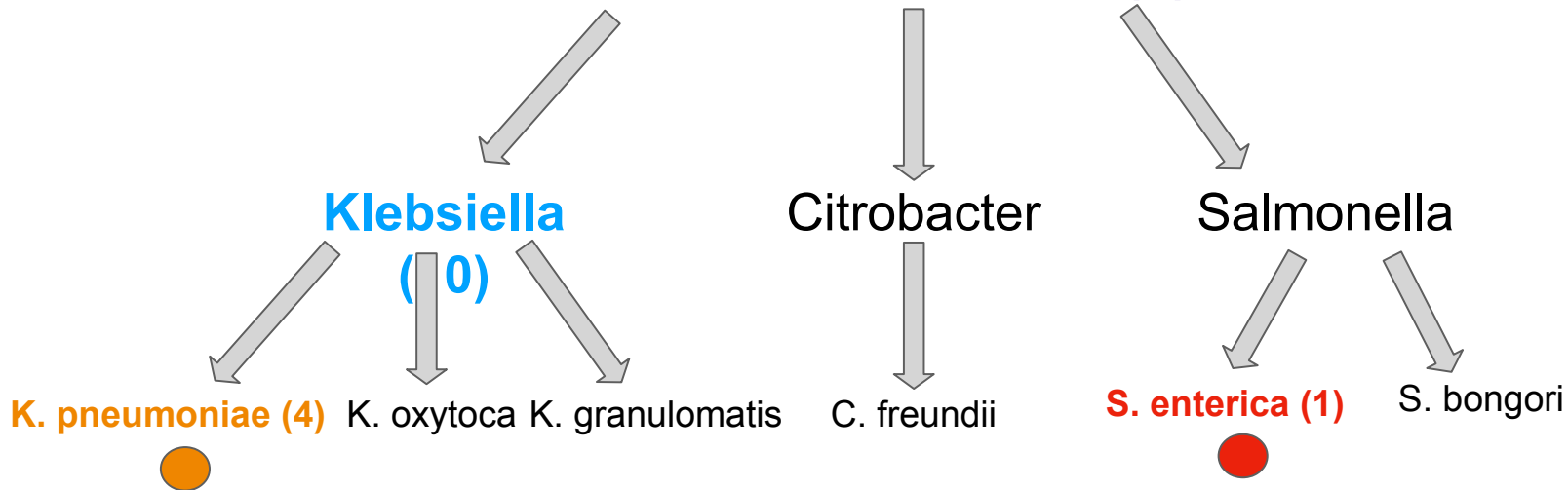


Taxonomy

LCA Mapping



Enterobacteriaceae (1)

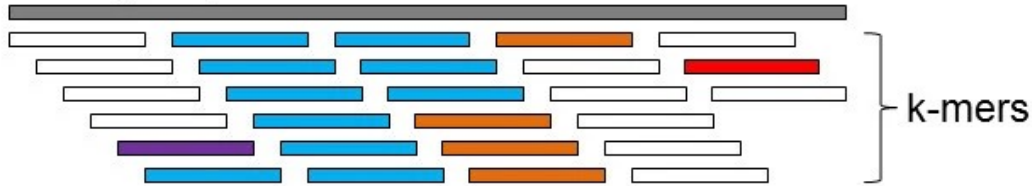


Taxonomy

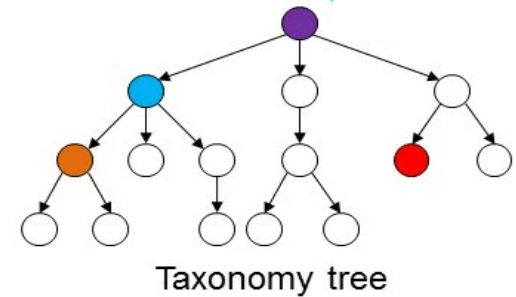
Taxonomic assignment: k-mer based

Example: [Kraken2](#)

1. Extract k-mers from sequence read (31-mers)



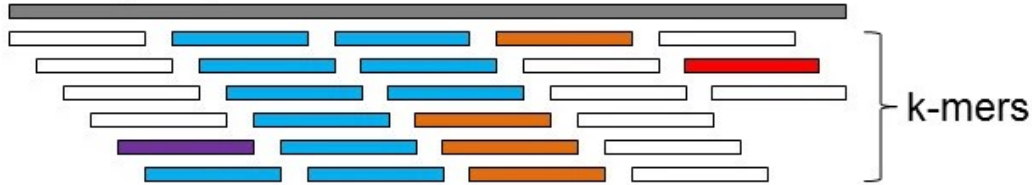
2. Compare to pre-computed database to determine taxonomic ID of reference sequences with specified k-mer (Lowest Common Ancestor, LCA, mapping)



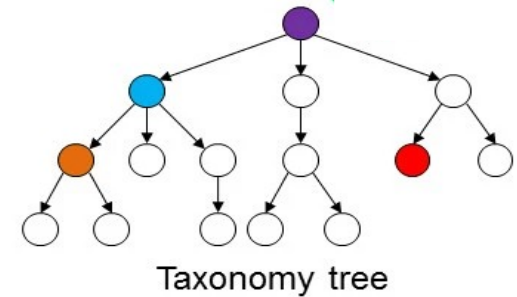
Taxonomic assignment: k-mer based

Example: [Kraken2](#)

1. Extract k-mers from sequence read (31-mers)

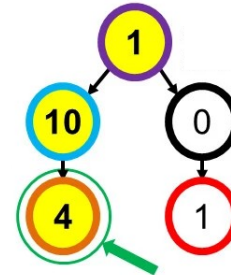


2. Compare to pre-computed database to determine taxonomic ID of reference sequences with specified k-mer (Lowest Common Ancestor, LCA, mapping)



3. Use subtree for classification. Find path with maximum sum and report taxonomic level for leaf

K. pneumoniae



Taxonomic assignment

Software to:

- Compare sequences against a reference database
 - k-mer based approaches (fast, simplify data processing to counting k-mers and comparing whether they are shared)
 - Alignment

Taxonomic assignment: Alignment

Alignment goal: Align unknown (query) sequences against a database of known sequences to identify the microbe of origin.

Database searching occurs through pairwise, local alignments against sequences found in the database

Taxonomic assignment: Alignment

Alignment goal: Query unknown sequences against a database of known sequences to identify the microbe of origin.

Database searching: Pairwise, local alignments against sequences found in the database

Global Alignment

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
   |||||      |||||  |||||      |||||
5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'
```

End-to-end

Local Alignment

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
      |||  |||||  |||||      |||||
5' TACTCACGGATGAGGTACTTTAGAGGC 3'
```

Substring

Taxonomic assignment: Alignment

Query: Unknown
sequence



Database of known
sequences

```
> Seq_1, unknown  
CTCATTCGGT
```

```
> species A  
GCCTCGATCTGTAGCCT
```

```
> species B  
ATGGACTCATTCCGGTCG
```

```
> species C  
TGCTAAATCGCGCCCTA
```

```
> species D  
TGGCTAAGTCATATAGC
```

Taxonomic assignment: Alignment



Query: Unknown
sequence



Database of known
sequences

> Seq_1, unknown
CTCATTTCGGT

> species A
GCCTCGATCTGTAGCCT

> species B
ATGGACTCATTTCGGTCG

> species C
TGCTAAATCGCGCCCTA

> species D
TGGCTAAGTCATATAGC

Sequence Alignment

> species A
GCCTCGATCTGTAGCCT
||| | ||
CTCATTTCGGT

> species B
ATGGACTCATTTCGGTCG
|||||||
CTCATTTCGGT

> species C
TGCTAAATCGCGCCCTA
|| | |||
CTCATTTCGGT

> species D
TGGCTAAGTCATATAGC
||| |
CTCATTTCGGT

Taxonomic assignment: Alignment

Query: Unknown
sequence



Database of known
sequences

```
> Seq_1, unknown
CTCATTCGGT
```

```
> species A
GCCTCGATCTGTAGCCT
```

```
> species B
ATGGACTCATTTCGGTCG
```

```
> species C
TGCTAAATCGCGCCCTA
```

```
> species D
TGGCTAAGTCATATAGC
```

Sequence Alignment

```
> species A
GCCTCGATCTGTAGCCT
```

```
|||  | ||
CTCATTCGGT
```

```
> species B
ATGGACTCATTTCGGTCG
```

```
|||||||
CTCATTCGGT
```

```
> species C
TGCTAAATCGCGCCCTA
```

```
|| | |||
CTCATTCGGT
```

```
> species D
TGGCTAAGTCATATAGC
```

```
||||  |
CTCATTCGGT
```

Seq_1 = Species B

Alignment Metrics

- Alignment length
 - How long is the alignment between the query and subject (sequence in the database)?

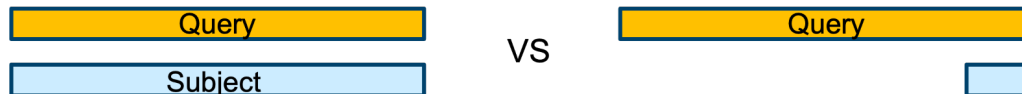
Alignment Metrics

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 - How long is the alignment between the query and subject (sequence in the database)?
- Query coverage
 - How much of my query sequence was covered by the alignment?



Alignment Metrics

- Alignment length
 - How long is the alignment between the query and subject (sequence in the database)?
- Query coverage
 - How much of my query sequence was covered by the alignment?



- Percent identity (% Id)
 - How similar is my query sequence to the subject sequence?

$$\% \text{ Id} = \frac{\text{\# identical nucleotides or amino acids}}{\text{Total \# nucleotides or amino acids in the alignment}} \times 100$$

Query GC**G**GGCGC**G**TTTCGCCAGGATCCATTG**G**CGA
|| ||||| |||||
Subject GC**A**GCGGC**T**TTTCGCCAGGATCCATTG**A**CGA

$$\% \text{ Id} = 27/30 * 100 = 90\%$$

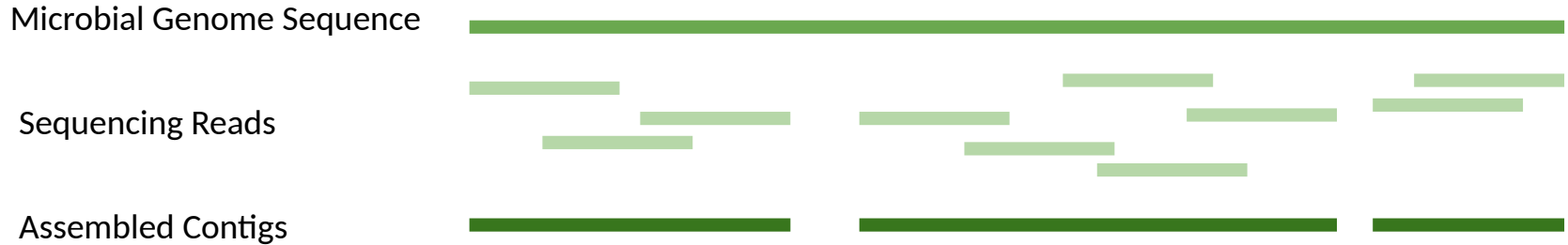
Taxonomic assignment: Alignment

Alignment goal: Align unknown sequences against a database of known sequences to identify the microbe of origin.

- **Query:**
 - Reads: Quality- and host- filtered sequence reads
 - Contigs: Assembled or merged sequence reads that form longer DNA sequences.

Contigs

Contigs: Assembled or merged sequence reads that form longer DNA sequences. The assembly aims to reconstruct the original sequence.



Sequence Assembly

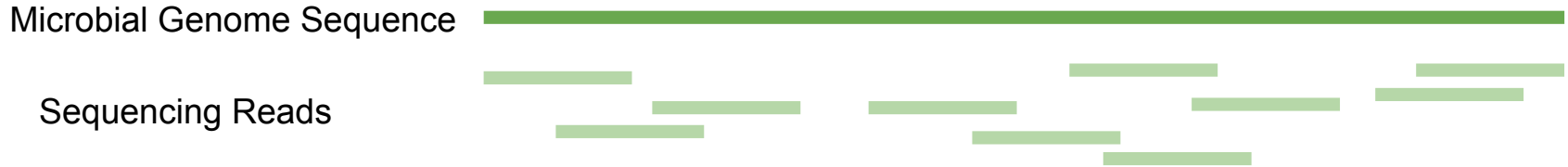
Assembly goal: Merge sequence reads into **contigs** to form longer DNA sequences and reconstruct the original sequence

Microbial Genome Sequence
(Sample)



Sequence Assembly

Assembly goal: Merge sequence reads into **contigs** to form longer DNA sequences and reconstruct the original sequence



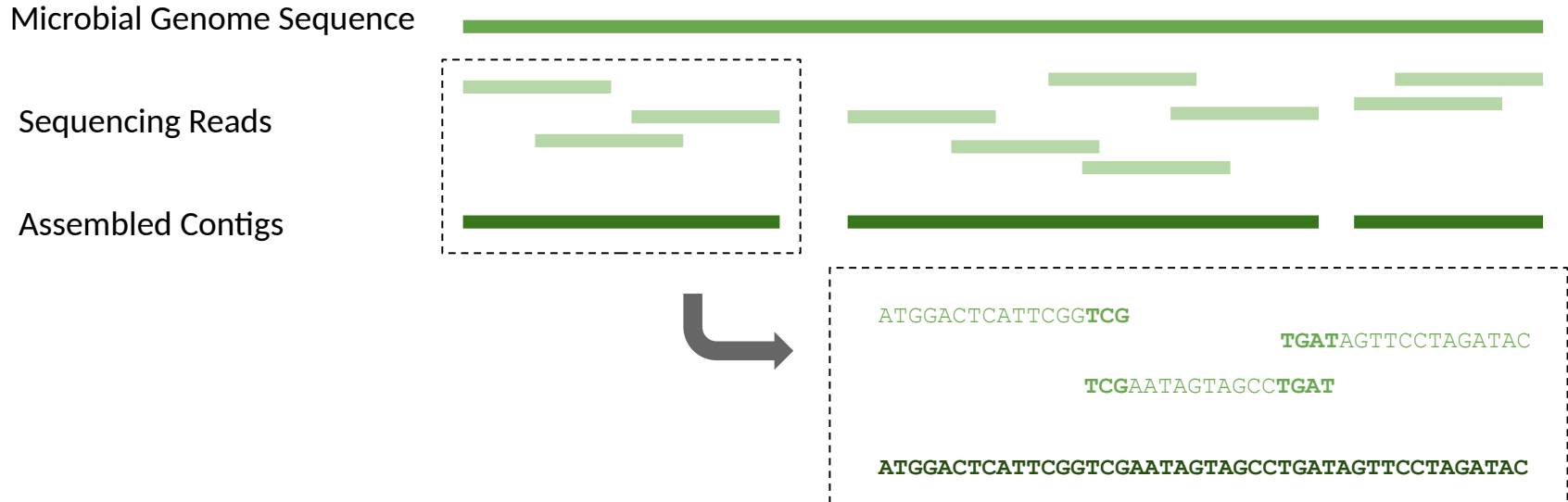
Sequence Assembly

Assembly goal: Merge sequence reads into **contigs** to form longer DNA sequences and reconstruct the original sequence



Sequence Assembly

Assembly goal: Merge sequence reads into **contigs** to form longer DNA sequences and reconstruct the original sequence



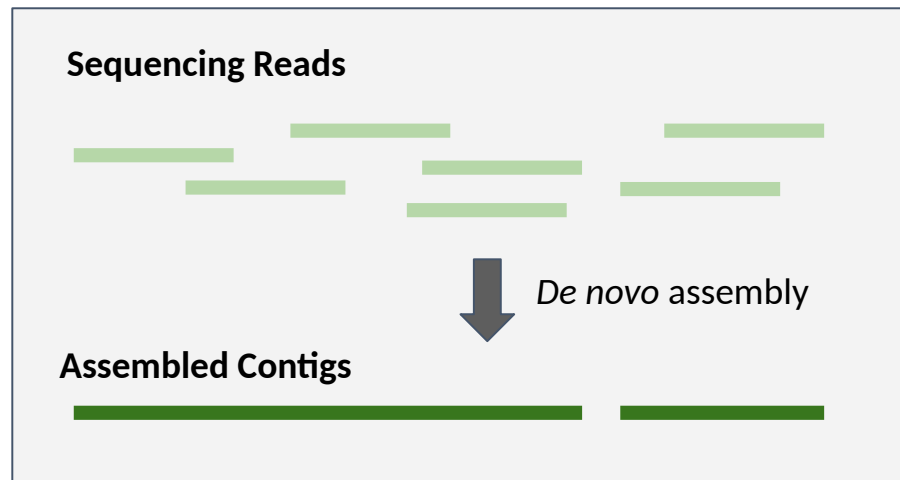
Assembly

Software: [SPAdes](#), [Flye](#), [MetaFlye](#)

Step: Contig assembly

Notes:

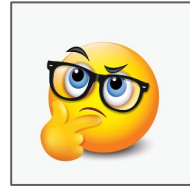
- **Input:** preprocessed reads in FASTA or FASTQ format
- Assembly choices based on the type of data (e.g., DNA vs RNA, WGS vs mNGS)



Taxonomic assignment: Alignment

Alignment goal: Query unknown sequences against a database of known sequences to identify the microbe of origin.

- **Query:** Reads or Contigs



If you could do anything (compute resources are not an issue), what would you use as a query?

Taxonomic assignment: Alignment

Alignment goal: Query unknown sequences against a database of known sequences to identify the microbe of origin.

- **Query:** Reads and/or Contigs
- **Database:** Any reference database of known organism sequences. Databases from the National Center for Biotechnology Information (NCBI) are ideal because they include comprehensive collections of publicly available sequences.

NCBI Databases

GenBank: Includes a collection of all publicly available DNA sequences. Researchers directly submit sequences to GenBank where the data goes through automated and manual processing to ensure data integrity (otherwise, the data is not curated).

RefSeq: Curated subset from GenBank. Curation ensures high accuracy and reliability.

NCBI Databases

	NT Database	NR Database
Definition	NT = nucleotide	NR = non-redundant protein

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What is in the database?	Publicly available nucleotide sequences (GenBank sequences except for sequences from NGS projects, such as WGS and SRA databases).	Amino acid sequences representing non-redundant, annotated coding regions from GenBank as well as records from protein databases (e.g., UniProt and PDB). Note: does not contain rRNA sequences given that these do not encode proteins

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How does it influence analysis?	NT alignments tend to be more accurate for most organisms (bacteria, eukaryotes, known viruses)	NR alignments are especially good for detecting divergent viruses , given that mutations in the NT sequence accumulate faster than the amino acid sequence

Amino Acid vs Nucleotide Alignments

Species 1 V I V A E L M A T E Y V L I V I A

100 % Id

Species 2 V I V A E L M A T E Y V L I V I A

Amino Acid vs Nucleotide Alignments

Species 1 V I V A E L M A T E Y V L I V I A

100 % Id

Species 2 V I V A E L M A T E Y V L I V I A

Sp. 1 GTG ATA GTA GCT GTC TTG ATC GTG ATC GCA TAT GAA CTG ATG GCT ACG GAA

Sp. 2 GTG ATA GTG GCC GTC TTG ATA GTG ATC GCG TAT GAA CTA ATG GCA ACG GAA

Amino Acid vs Nucleotide Alignments

Species 1 V I V A E L M A T E Y V L I V I A

100 % Id

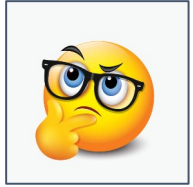
Species 2 V I V A E L M A T E Y V L I V I A

88 % Id

Sp. 1 GTG ATA GT^A GC^T GTC TTG AT^C GTG ATC GC^A TAT GAA CT^G ATG GC^T ACG GAA

Sp. 2 GTG ATA GT^G GC^C GTC TTG AT^A GTG ATC GC^G TAT GAA CT^A ATG GC^A ACG GAA

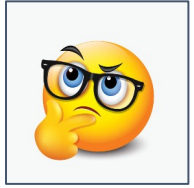
Where would you see the following *E. coli* matches?



E. coli **16S rRNA**

	NT Database	NR Database
Match (or hit) to <i>E. coli</i> based on...		

Where would you see the following *E. coli* matches?



E. coli **polymerase**

	NT Database	NR Database
Match (or hit) to <i>E. coli</i> based on...		

Where would you see the following *E. coli* matches?

E. coli polymerase

E. coli 16S rRNA

	NT Database	NR Database
Match (or hit) to <i>E. coli</i> based on...	16S rRNA Polymerase	Polymerase

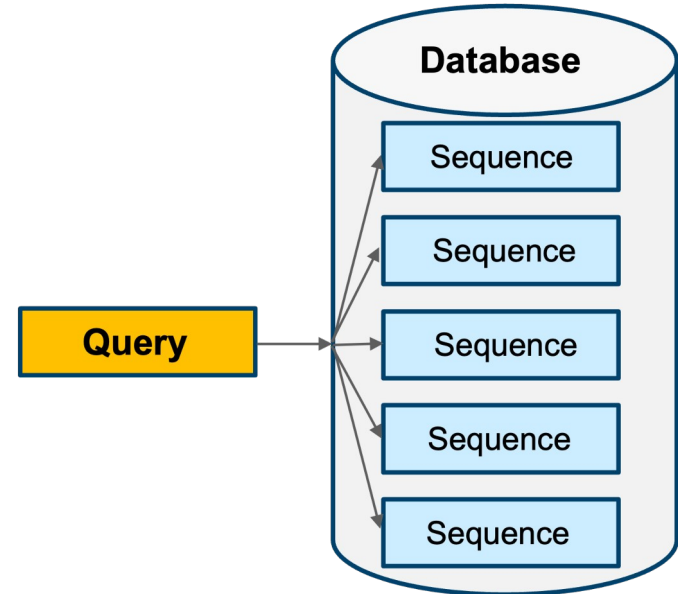
Taxonomic assignment

Example software: BLAST, minimap2, Kraken2, DIAMOND

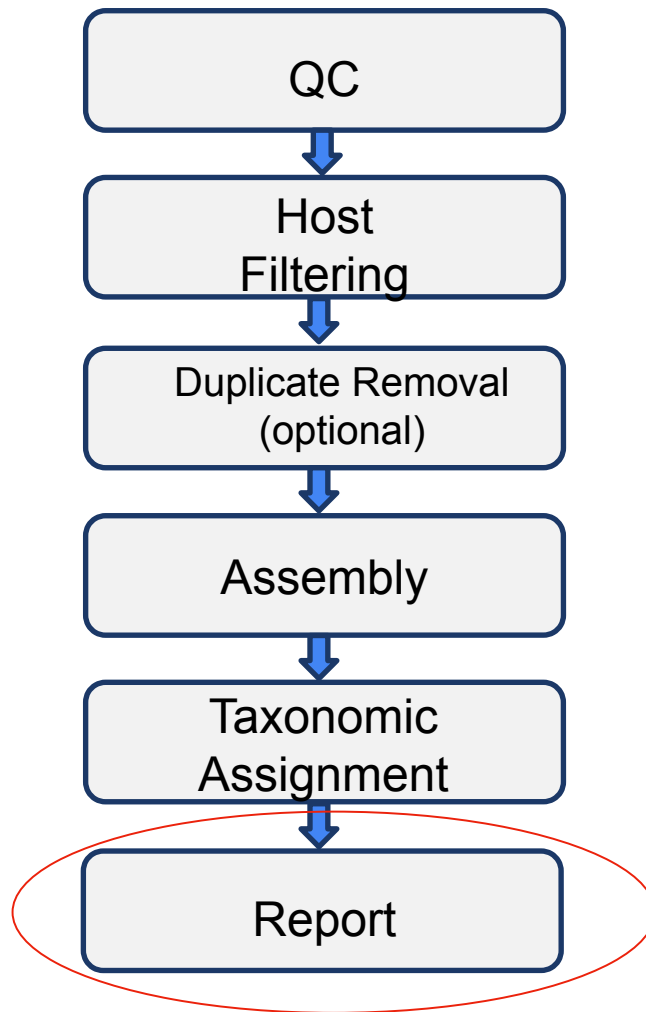
Step: Querying reference database

Notes:

- BLAST – aligns nucleotide and amino acid sequences
- Minimap2 – aligns nucleotide sequences
- Kraken2 - kmer-based approach for searching nucleotide and amino acid sequences
- DIAMOND – aligns amino acid sequences



Common mNGS Pipeline Steps



Basic Local Alignment Search Tool (BLAST)

- Gold standard tool for sequence similarity searching against databases (sensitive!)

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- Gold standard tool for sequence similarity searching against databases (sensitive!)
- Provides alignment metrics to evaluate the significance of alignments
 - **Bit score:** Metric derived from the raw alignment score. The higher the bit score, the more significant the alignment (higher similarity).
 - **E-value:** Number of matches you expect to see by chance. The lower the E-value, the more significant the alignment.

Basic Local Alignment Search Tool (BLAST)

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 - Bit score
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- Great resource for genomic applications
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 - Identifying unknown sequences (taxonomy)
 - Genome annotation

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 - Identifying unknown sequences (taxonomy)
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- Limitation: Time required for alignment scales with query and database length

Let's BLAST!

Activity: BLAST

